



THE APERTURE METHOD™ · CLASSROOM PROGRAM

# Answer Key

What each card is really about, and where to push

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*The question this answers: **Where does the room need to be pushed?***

PREPARED FOR

**Grades 8–12**

The Woodlands, Texas

Facilitator copy · Edition One · The Aperture Method™

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**A NOTE ON THE BUSINESSES:** Every business in this packet is invented. The streets, parks and shopping centers are real places in The Woodlands, Texas, so the exercise feels like home, but no real business is described, and no real numbers are used.

## FACILITATOR ONLY

## What each card is really about

Each business is broken in a different way, so no two teams reach the same answer and the share-out never repeats itself.

| # | Business                | Lens              | The bottleneck in six words      | Worth per year |
|---|-------------------------|-------------------|----------------------------------|----------------|
| 1 | Waterway Waffle Co.     | THE THING         | One waffle iron caps the rush    | ~\$25,000      |
| 2 | Creekside Cuts          | WHO               | Two in three never come back     | ~\$23,400      |
| 3 | Panther Creek Lawn Crew | MONEY             | The price has not moved in years | ~\$4,200       |
| 4 | Alden Bridge Bakehouse  | WHO DOES THE WORK | Only Maria can decorate cakes    | ~\$12,000      |
| 5 | Town Green Smoothie Bar | WHERE & WHEN      | Closed on the nights that matter | ~\$37,000      |
| 6 | Kickback Sneakers       | WHO               | Excellent shop, nobody knows it  | ~\$9,000       |

### Hold the line on one answer

Teams will want to name two or three problems. Every business has more than one thing wrong with it; the skill is deciding which one to fix first. Make them choose. That single act of subtraction is the most transferable thing in the hour.

## CARD 1 · LENS: THE THING

# Waterway Waffle Co.

## The bottleneck

One waffle iron. The machine is the ceiling.

## Why

- 4 minutes a waffle means 15 waffles an hour, absolute maximum.
- The rush is 2 hours long.  $15 \times 2 = 30$  waffles is all they can physically produce when the line is longest.
- They are not losing customers because the waffles are bad, or because nobody knows about them, or because the price is wrong. All of those are fine. They are losing customers because there is one iron.

## The arithmetic

|                           | Working           | Result                               |
|---------------------------|-------------------|--------------------------------------|
| 40 people walked away     | $\times \$8$ each | = \$320 of sales lost in one morning |
| Profit on a waffle        | $\$8 - \$2$       | = \$6                                |
| Lost profit, one Saturday | $40 \times \$6$   | = \$240                              |
| Saturday and Sunday       | $\$240 \times 2$  | = \$480 a weekend                    |
| Every weekend, all year   | $\$480 \times 52$ | = about \$25,000 a year              |

## The fix

A second waffle iron costs about \$600. It pays for itself before the end of the second weekend, and then it makes about \$25,000 a year, forever.

## Push here

Sharp teams will notice a second iron needs a second pair of hands. During the rush the order-taker can cook. That is a schedule change, not a hire. Let them find it. It is a real objection and they are right to raise it.

## CARD 2 · LENS: WHO

## Creekside Cuts

## The bottleneck

Two out of every three new customers never return. The shop is a bucket with a hole in it.

## Why

- The shop is not bad at getting customers. It got 2,400 of them. It is bad at keeping them.
- Money spent on ads pours new people into a bucket that leaks 65% of them straight back out.
- A customer who returns is not a little more valuable than one who does not. They are six times more valuable.

## The arithmetic

|                                     | Working                                                        | Result            |
|-------------------------------------|----------------------------------------------------------------|-------------------|
| A one-time customer is worth        | $1 \times \$30$                                                | = \$30            |
| A repeat customer is worth          | $6 \times \$30$                                                | = \$180           |
| So a repeat customer is worth       | $\$180 \div \$30$                                              | = 6 times as much |
| Ads cost                            | $\$900 \times 12 \text{ months}$                               | = \$10,800 a year |
| Win back just 10% of the one-timers | $156 \text{ people} \times 5 \text{ extra visits} \times \$30$ | = \$23,400 a year |
| Extra ad money needed               |                                                                | = \$0             |

## The fix

Book the next appointment before the customer leaves the chair. Text them at week five. Neither costs a dollar. Both are worth more than doubling the ad budget.

## Push here

Someone will say "spend more on ads." Ask them: if you pour water faster into a bucket with a hole in it, what happens? This is the moment the whole lesson lands. Do not rush it.

## CARD 3 · LENS: MONEY

## Panther Creek Lawn Crew

## The bottleneck

The price is wrong. They are working a full day for less than minimum wage, and it is nobody's fault but the price tag.

## Why

- Every customer says yes immediately, and there is a waiting list. Both of those are signs the price is too low, not that the business is good.
- Doing MORE yards at \$35 does not fix it. It just means more hours at \$5.50.
- This is the hardest idea in the whole lesson: being busy and making money are different things.

## The arithmetic

|                              | Working                                           | Result                                                 |
|------------------------------|---------------------------------------------------|--------------------------------------------------------|
| Revenue                      | 8 yards $\times$ \$35                             | = \$280                                                |
| Minus supplies               | 8 $\times$ \$9                                    | = -\$72                                                |
| Minus Marcus                 | 8 hours $\times$ \$15                             | = -\$120                                               |
| Left for Dev and Ray         |                                                   | = \$88                                                 |
| Each, per hour               | $\$88 \div 2 \text{ people} \div 8 \text{ hours}$ | = \$5.50 an hour                                       |
| — —                          |                                                   |                                                        |
| Now raise the price to \$45  | $8 \times \$45 = \$360 - \$72 - \$120$            | = \$168                                                |
| Each, per hour               | $\$168 \div 2 \div 8$                             | = \$10.50 an hour                                      |
| If they LOSE one customer    | $7 \times \$45 = \$315 - \$63 - \$105$            | = \$147                                                |
| Each, per hour (7-hour day)  | $\$147 \div 2 \div 7$                             | = \$10.50 an hour, and home an hour early              |
| Extra profit, every Saturday | $\$168 - \$88$                                    | = \$80                                                 |
| Across a year of Saturdays   | $\$80 \times 52$                                  | = about \$4,200 a year, and their pay per hour doubles |

## The fix

Charge \$45. If one customer leaves, they earn exactly the same money per hour and get an hour of their Saturday back. The waiting list fills the gap by lunchtime.

**Push here**

Push on the last row. Same money, fewer customers, more free time. Ask the room: is that a worse business or a better one? Most kids have never been given permission to think a smaller number can be the right answer.

## CARD 4 · LENS: WHO DOES THE WORK

# Alden Bridge Bakehouse

## The bottleneck

One person is the only one who can do the most valuable thing in the business, and she spends half her day doing things anyone could do.

## Why

- Maria's hour is the scarcest thing the bakery owns, and she spends it on a register that a \$14-an-hour employee could run.
- The business does not have a cake problem. It has a Maria problem.
- This is the single most common thing that holds back a real small business, and almost nobody who owns one can see it from the inside.

## The arithmetic

|                                         | Working             | Result                                    |
|-----------------------------------------|---------------------|-------------------------------------------|
| Profit on one cake                      | $\$95 - \$22$       | $= \$73$                                  |
| Maria's time per cake                   | 3 hours             |                                           |
| So an hour of Maria decorating is worth | $\$73 \div 3$       | $= \text{about } \$24$                    |
| Profit on one pastry                    | $\$4 - \$1.20$      | $= \$2.80$                                |
| Pastries needed to equal one cake       | $\$73 \div \$2.80$  | $= 26 \text{ pastries}$                   |
| Cakes turned away last month            | $14 \times \$73$    | $= \$1,022 \text{ lost}$                  |
| Every month, all year                   | $\$1,022 \times 12$ | $= \text{about } \$12,000 \text{ a year}$ |

## The fix

Teach one employee to decorate. Or take the register away from Maria. Either one costs almost nothing and buys back the most expensive hours in the building.

## Push here

Ask what happens the week Maria gets the flu. The room will figure out on their own that the business does not just slow down. A whole product line stops.

## CARD 5 · LENS: WHERE &amp; WHEN

## Town Green Smoothie Bar

**The bottleneck**

They have one of the best locations in Montgomery County and they are closed at the exact hours it is worth something.

**Why**

- Location is not just a place. It is a place AND a time.
- Nothing about the shop needs to change. Not the product, not the price, not the sign. Just the hours on the door.
- Notice the hiring sign: the thing standing between them and the money is that nobody has bothered to make a phone call.

**The arithmetic**

|                                     | Working                        | Result                  |
|-------------------------------------|--------------------------------|-------------------------|
| Concertgoers on one night           |                                | 12,000 people           |
| If only 1 in 100 buys a smoothie    | $12,000 \div 100$              | = 120 smoothies         |
| Sales that night                    | $120 \times \$9$               | = \$1,080               |
| Minus ingredients                   | $120 \times \$2.70$            | = -\$324                |
| Minus two extra staff, 5 hours each | $10 \text{ hours} \times \$14$ | = -\$140                |
| Profit, one concert night           |                                | = \$616                 |
| Across 60 concerts a year           | $\$616 \times 60$              | = about \$37,000 a year |

**The fix**

Open on concert nights. That is the entire recommendation. It costs a schedule change and two phone calls to people who already applied.

**Push here**

Ask: 1 in 100. Is that optimistic or pessimistic? Most rooms guess it is optimistic. It is not. Let them argue about it. Arguing about an assumption is exactly what real analysts do.



## CARD 6 · LENS: WHO

# Kickback Sneakers

**The bottleneck**

The business is good and invisible. Seven thousand of its exact customers live down the road and do not know it exists.

**Why**

- Every other lens comes back clean. The product is right, the price works, the people are good, the hours are fine.
- When four lenses are clean and the numbers are still small, the answer is usually that not enough people know.
- This is the ONLY card in the set where "market it more" is the right answer. That is deliberate.

**The arithmetic**

|                                          | Working             | Result                              |
|------------------------------------------|---------------------|-------------------------------------|
| Profit today                             | 45 pairs × \$50     | = \$2,250 a month                   |
| Say posting brings 15 more pairs a month | 15 × \$50           | = \$750 a month                     |
| Across a year                            | \$750 × 12          | = \$9,000 a year                    |
| Cost of an Instagram account             |                     | = \$0                               |
| Cost in time                             | 20 minutes a day    | = about 120 hours a year            |
| So the owner earns                       | \$9,000 ÷ 120 hours | = \$75 an hour for posting sneakers |

**The fix**

Post. Every day. And put a sign at the bottom of the stairs.

**Push here**

Save this card for last in the share-out. After five cards where "advertise more" was the wrong answer, the room will have learned to distrust it, and then here it is right. That whiplash is the point: the answer depends on the evidence, not on the reflex.

## THE POINT OF THE WHOLE THING

### If they only take one thing

Five of these six businesses would be made worse, not better, by more customers. The waffle shop already turns people away. The lawn crew loses money faster with every extra yard. The barbershop pours new customers into a bucket with a hole in it. The bakery cannot make what it already sold. The smoothie bar is closed.

Only Kickback Sneakers, a genuinely good business that nobody has heard of, is fixed by more attention. One card in six.

**"More customers" is the first answer almost everyone gives, and it is right about one time in six. That ratio is worth saying out loud before the bell.**